EC2L104 – Introduction to EDA Tools

LTspice Exercise Report

MOSFET Characterisation, Parametric Study and CMOS Inverter
using the 65 nm PTM Model Card

Name: Tuladhar Reddy Madireddi

Roll No.: 25EC01048

GitHub repo: _____ (`LTSPICE`)

Contents

1 S1 – Reading the Model Card **2**
1.1 Extracted Card Parameters . . . 2
1.2 Calculations . . . 2
1.3 .op Results at $V_{GS} = V_{DS} = 1.1$ V . . . 2

2 S2 – Transfer Characteristics **3**

3 S3 – Output Characteristics **4**

4 S4 – Physics Check **5**

5 S5 – Parametric Variation **6**
5.1 Experiment 1: VTH0 . . . 6
5.2 Experiment 2: TOXE . . . 7
5.3 Experiment 3: L (drawn) . . . 7
5.4 Experiment 4: XL . . . 8
5.5 Experiment 5: LINT . . . 8
5.6 Experiment 6: W (drawn) . . . 9
5.7 Experiment 7: WINT . . . 9
5.8 Experiment 8: NDEP . . . 10
5.9 Experiment 9: NSD . . . 10
5.10 Experiment 10: XJ . . . 11
5.11 Experiment 11: NGATE . . . 11
5.12 Experiment 12: U0 . . . 12
5.13 Experiment 13: VSAT . . . 12
5.14 Experiment 14: RDSW . . . 13

6 S6 – Approximate Match to TCAD **14**

7 S7 – CMOS Inverter **15**
7.1 Circuit and Parameters . . . 15
7.2 Voltage Transfer Characteristic . . . 15

1 S1 – Reading the Model Card

Circuit / Testbench

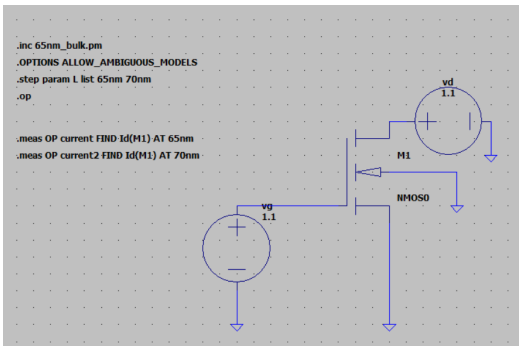


Figure 1: LTspice testbench for the `.op` runs ($L = 65$ nm and 70 nm).

```
.step param L list 65nm 70nm
.op
.meas OP current FIND Id(M1) AT 65nm
.meas OP current2 FIND Id(M1) AT 70nm
```

1.1 Extracted Card Parameters

Parameter	Value
XL (nm)	-30
LINT (nm)	5.25
WINT (nm)	5
TOXE (nm)	1.85

Table 1: Values read from the model card.

1.2 Calculations

$$L_{eff} = L_{drawn} + XL - 2\,LINT = L_{drawn} - 40.5 \text{ nm}$$
$$W_{eff} = W_{drawn} - 2\,WINT = W_{drawn} - 10 \text{ nm}$$
$$C_{ox} = \frac{3.9\,\varepsilon_0}{TOXE} = 1.87\ \mu\text{F/cm}^2$$

L (nm)	L_{eff} (nm)
65	24.5
70	29.5

Table 2: Computed effective channel length.

1.3 .op Results at $V_{GS} = V_{DS} = 1.1$ V

L (nm)	I_D (mA)
65	1.15
70	1.05

Table 3: Drain current from `.op`.

Observation: We can see that with increase in channel length a proportional decrease in drain current. $\frac{L_{70}}{L_{65}} = 1.08$, $\frac{I_{D65}}{I_{D70}} = \frac{1.15}{1.05} = 1.09$, hence $\frac{L_{70}}{L_{65}} \approx \frac{I_{D65}}{I_{D70}}$.

Conclusion: From the observation we can conclude that I_D **varies inversly proportional to channel length** which matches to the theory we learnt, $I_D \propto \frac{1}{L}$.

2 S2 – Transfer Characteristics

Circuit / Testbench

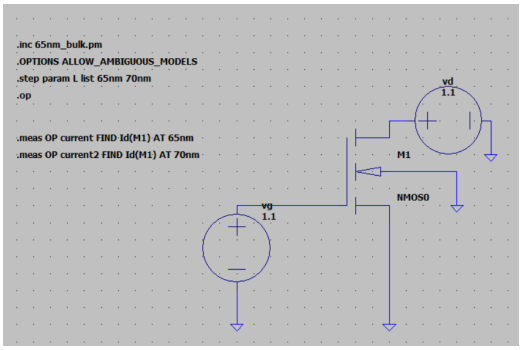


Figure 2: LTspice testbench for I_D–V_G (V_{DS} = 50 mV and 1.1 V).

```
.dc vg 0 1.1 0.01
.step param v list 0.05 1.1
```

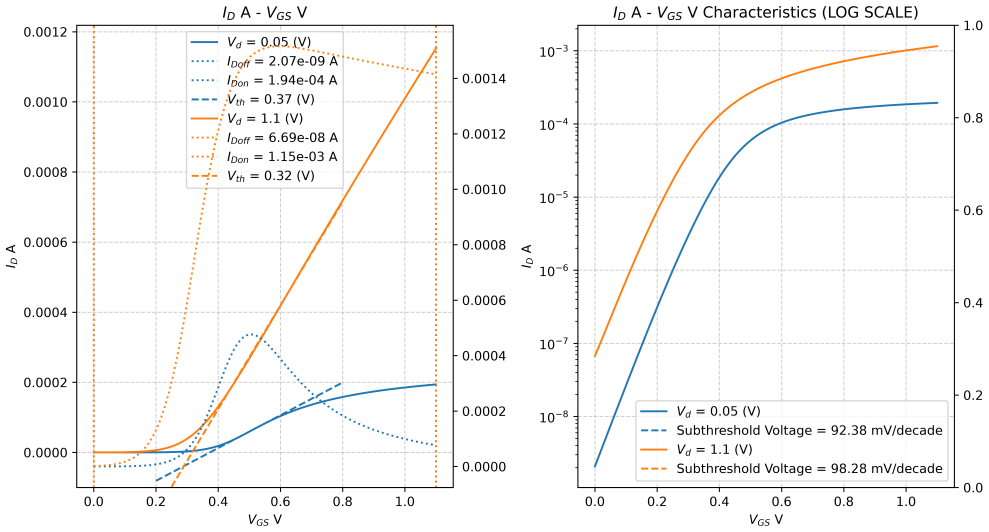


Figure 3: I_D–V_G at V_{DS} = 50 mV and 1.1 V (L = 65 nm).

Parameter	Method	Extracted (SPICE)	Table 5.1
V_{TH} (V)	Tangent at maximum gm, VDS = 50 mV, intercept with I_D =0	0.37	0.38
I_{ON} (mA/μm)	$V_{GS} = V_{DS}$ = 1.1 V	1.15	1.15
I_{OFF} (nA/μm)	V_{GS} = 0, V_{DS} = 1.1 V	66.9	67
SS (mV/dec)	Extrapolation similar to V_{th}	92.38 (0.05 V), 98.28 (1.1 V),	90 (50 mV), 96(1.1 V)

Table 4: Comparison of extracted parameters

Observation: The extracted V_{TH}, I_{ON}, I_{OFF} and SS agree with Table 5.1 to within about 3%, and the higher V_{DS} curve shows a larger SS (98.28 vs 92.38 mV/dec) and a larger I_{OFF} because of drain-induced barrier lowering.

Conclusion: The 65 nm model card reproduces the reference transfer characteristics, with $V_{TH} \approx$ 0.37 V, I_{ON} = 1.15 mA/μm and $I_{OFF} \approx$ 67 nA/μm.

3 S3 – Output Characteristics

Circuit / Testbench

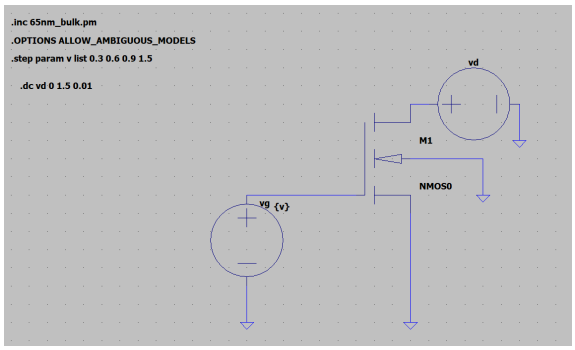
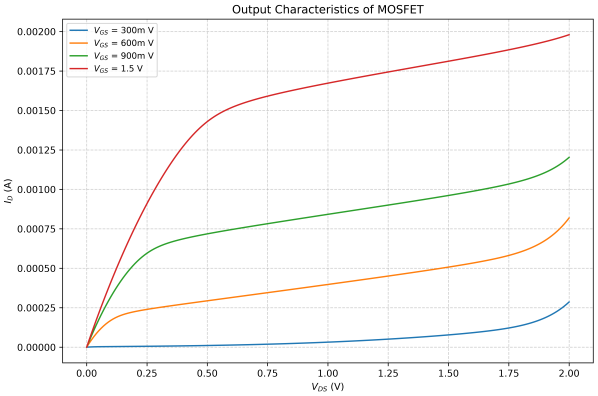
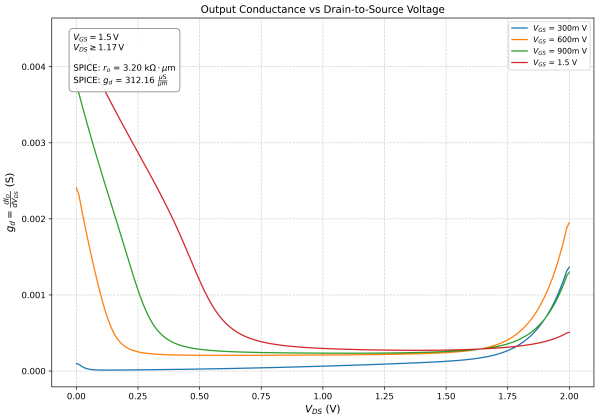


Figure 4: LTspice testbench for the I_D–V_D family.

```
.dc vd 0 1.5 0.01
.step param v list 0.3 0.6 0.9 1.5
```



(a) I_D–V_D family at $V_{GS} = 0.3, 0.6, 0.9, 1.5$ V



(b) g_{ds} (via `np.gradient`) versus V_{DS}; saturation region marked at $V_{GS} = 1.5$ V

Quantity (at $V_{GS} = 1.5$ V, saturation)	SPICE
V_{DS} range used for g_{ds} (V)	1.17 - 2
g_{ds} (μS/μm)	312.16
$r_o = 1/g_{ds}$ (k$\Omega \cdot \mu$m)	3.20

Table 5: Output conductance and resistance.

Observation: Each curve rises linearly in the triode region and then flattens in saturation, but the saturation current keeps a small positive slope, giving $g_{ds} = 312.16$ μS/μm.

Conclusion: The device has a finite output resistance $r_o = 3.20$ k$\Omega \cdot \mu$m caused by channel-length modulation and DIBL, so it is not an ideal current source.

4 S4 – Physics Check

Circuit / Testbench

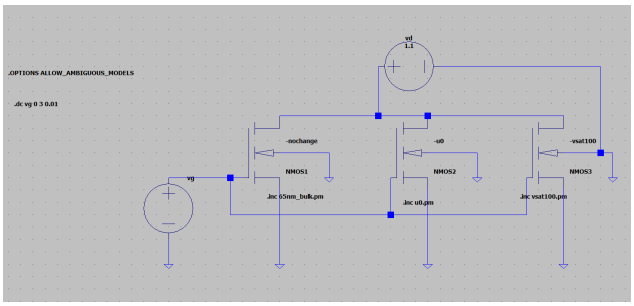


Figure 6: LTspice testbench for the model-parameter checks (UA, UB, $VSAT$).

```
.dc vg 0 3 0.01
```

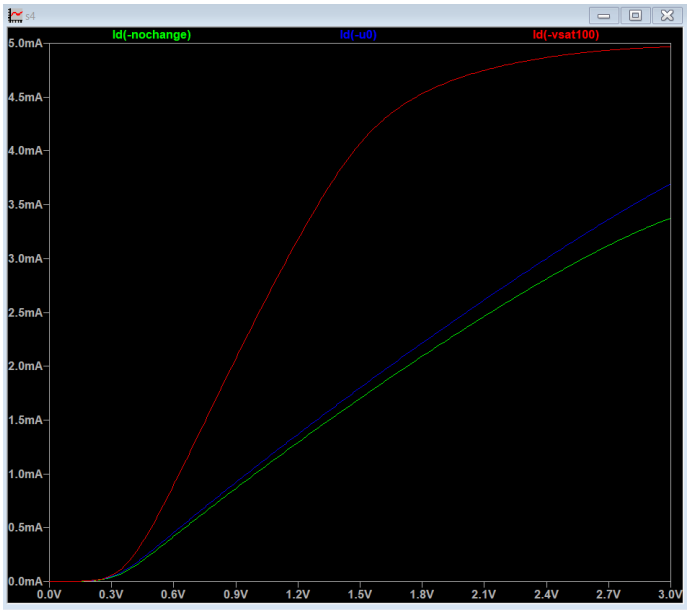


Figure 7: I_D–V_G overlay: nominal, $UA = UB = 0$, and $VSAT \times 100$.

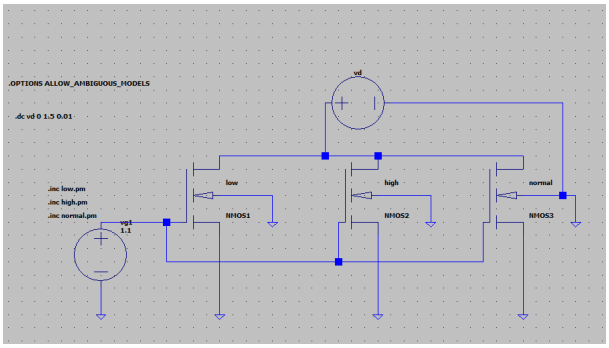
$UA = UB = 0 \leftrightarrow$ **TCAD *Enormal* (mobility degradation) model:** A large vertical gate field pulls carriers against the Si–SiO_2 interface where surface-roughness and phonon scattering reduce the effective mobility, and UA and UB model this reduction; setting them to zero removes the degradation, so I_D keeps growing with V_{GS} and I_{ON} becomes noticeably higher than nominal, while V_{TH} and I_{OFF} stay essentially unchanged.

$VSAT \times 100 \leftrightarrow$ **TCAD *HighFieldSaturation* model:** In a short channel the lateral field is high enough for carriers to reach the saturation velocity v_{sat}, which caps the current at roughly $WC_{ox}(V_{GS} - V_{TH})v_{sat}$; raising VSAT by 100× effectively removes this limit, so the drain current follows the mobility-limited (quadratic) law, I_{ON} increases strongly and the output curves no longer saturate early.

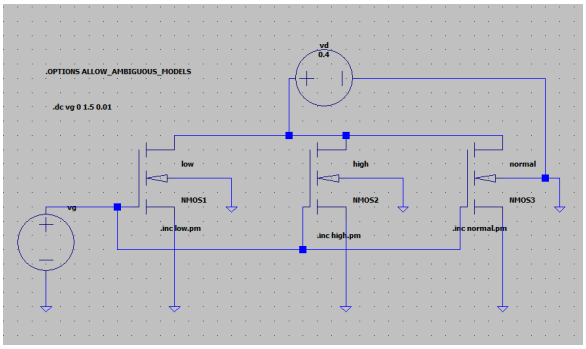
Conclusion: Velocity saturation limits I_{ON} the most in a 65 nm device, because the lateral field is very high, with vertical-field mobility degradation as a secondary limit.

5 S5 – Parametric Variation

Circuit / Testbench



(a) Circuit of I_D-V_G



(b) Circuit of I_D-V_D

```
.dc vg 0 1.5 0.01
```

```
.dc vd 0 1.5 0.01
```

Each experiment varies one parameter around its nominal value. I_D–V_G is simulated at $V_{DS} = 0.4$ V and I_D–V_D at $V_{GS} = 1.1$ V.

5.1 Experiment 1: VTH0

Values (low / nominal / high): 0.30 / 0.423 / 0.55 V

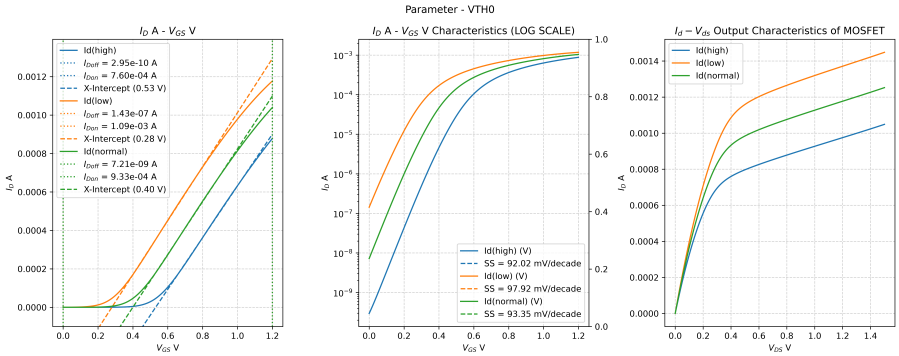


Figure 9: Exp. 1: effect of varying VTH0 (left: I_D–V_G with V_{TH}; middle: log scale with SS; right: I_D–V_D).

Case	V_{TH} (V)	SS (mV/dec)	I_{ON} (μA/μm)	I_{OFF} (nA/μm)
Low	0.28	97.92	1090	143
Nominal	0.40	93.35	933	7.21
High	0.53	92.02	760	0.295

Observation: As expected got V_{th} close to the specified values in pm file

5.2 Experiment 2: TOXE

Values (low / nominal / high): 1.5 / 1.85 / 2.5 nm

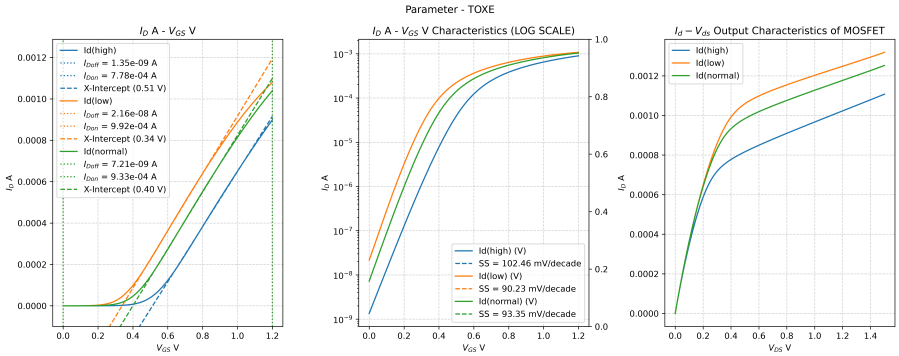


Figure 10: Exp. 2: effect of varying TOXE (left: I_D–V_G with V_{TH}; middle: log scale with SS; right: I_D–V_D).

Case	V_{TH} (V)	SS (mV/dec)	I_{ON} (μA/μm)	I_{OFF} (nA/μm)
Low	0.34	90.23	992	21.6
Nominal	0.40	93.35	933	7.21
High	0.51	102.46	778	1.35

Observation: A thicker oxide lowers C_{ox} and weakens gate control, so V_{TH} rises (0.34 to 0.51 V), I_{ON} falls (992 to 778 μA/μm), I_{OFF} drops and SS degrades.

5.3 Experiment 3: L (drawn)

Values (low / nominal / high): 55 / 65 / 100 nm

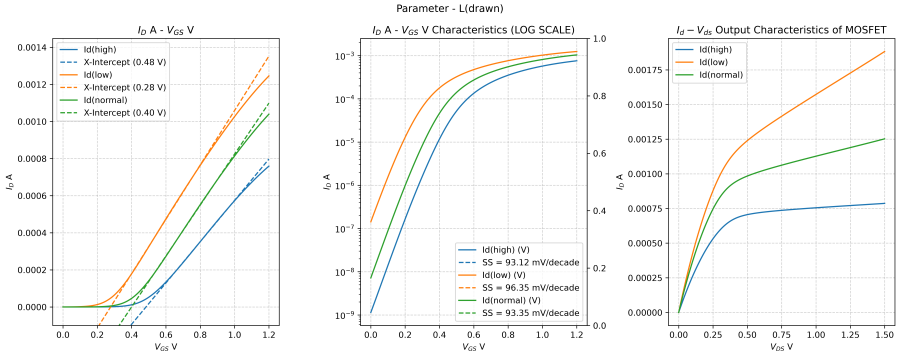


Figure 11: Exp. 3: effect of varying L (drawn) (left: I_D–V_G with V_{TH}; middle: log scale with SS; right: I_D–V_D).

Case	V_{TH} (V)	SS (mV/dec)	I_{ON} (μA/μm)	I_{OFF} (nA/μm)
Low	0.28	96.35	≈1200	≈140
Nominal	0.40	93.35	933	7.21
High	0.48	93.12	≈700	≈1

Observation: A shorter channel gives a lower V_{TH} (short-channel effect), a higher I_{ON} (≈1200 vs ≈700 μA/μm) and a much larger I_{OFF}.

5.4 Experiment 4: XL

Values (low / nominal / high): -35 / -30 / -15 nm

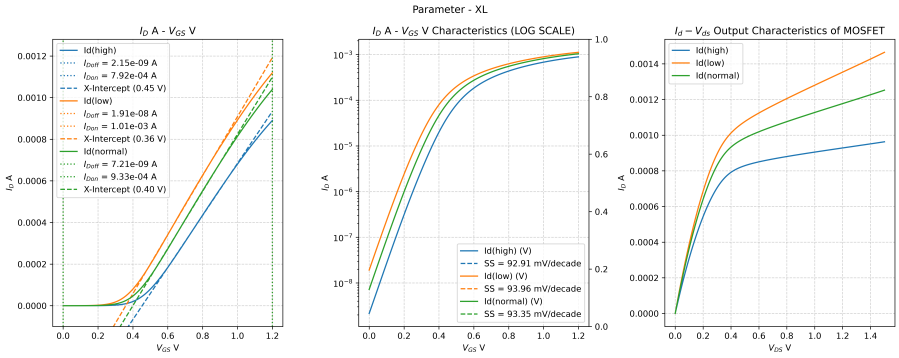


Figure 12: Exp. 4: effect of varying XL (left: I_D–V_G with V_{TH}; middle: log scale with SS; right: I_D–V_D).

Case	V_{TH} (V)	SS (mV/dec)	I_{ON} (μA/μm)	I_{OFF} (nA/μm)
Low	0.36	93.96	1010	19.1
Nominal	0.40	93.35	933	7.21
High	0.45	92.91	792	2.15

Observation: A more negative XL shortens L_{eff}, so V_{TH} falls, I_{ON} rises and I_{OFF} increases, the same trend as reducing the drawn L.

5.5 Experiment 5: LINT

Values (low / nominal / high): 2 / 5.25 / 8 nm

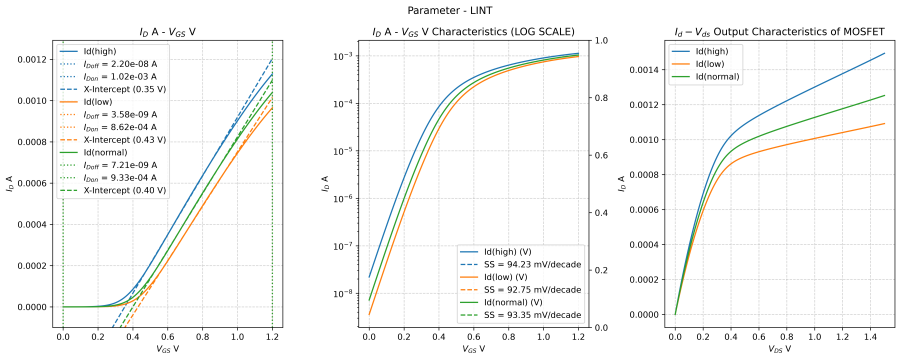


Figure 13: Exp. 5: effect of varying LINT (left: I_D–V_G with V_{TH}; middle: log scale with SS; right: I_D–V_D).

Case	V_{TH} (V)	SS (mV/dec)	I_{ON} (μA/μm)	I_{OFF} (nA/μm)
Low	0.43	92.75	862	3.58
Nominal	0.40	93.35	933	7.21
High	0.35	94.23	1020	22.0

Observation: A larger LINT shortens $L_{eff} = L + XL - 2\,LINT$, so V_{TH} drops (0.43 to 0.35 V) while I_{ON} and I_{OFF} rise.

5.6 Experiment 6: W (drawn)

Values (low / nominal / high): 0.5 / 1 / 2 μm

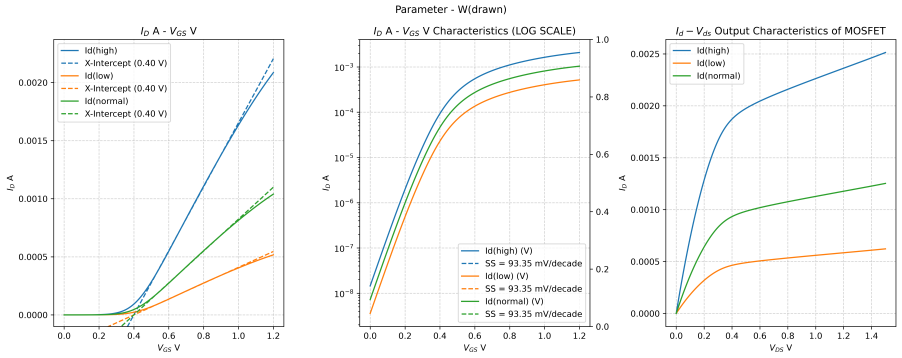


Figure 14: Exp. 6: effect of varying W (drawn) (left: I_D–V_G with V_{TH}; middle: log scale with SS; right: I_D–V_D).

Case	V_{TH} (V)	SS (mV/dec)	I_{ON} (μA/μm)	I_{OFF} (nA/μm)
Low	0.40	93.35	933	7.21
Nominal	0.40	93.35	933	7.21
High	0.40	93.35	933	7.21

Observation: All parameters are identical for the three widths because they are normalised per μm, so width only scales the absolute current linearly.

5.7 Experiment 7: WINT

Values (low / nominal / high): 0 / 5 / 40 nm

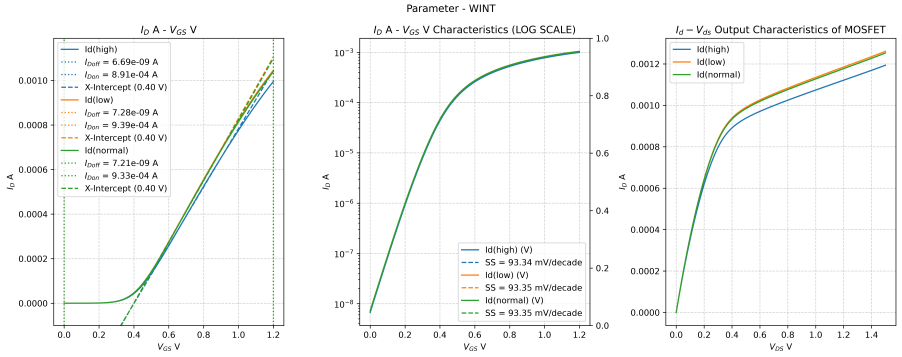


Figure 15: Exp. 7: effect of varying WINT (left: I_D–V_G with V_{TH}; middle: log scale with SS; right: I_D–V_D).

Case	V_{TH} (V)	SS (mV/dec)	I_{ON} (μA/μm)	I_{OFF} (nA/μm)
Low	0.40	93.35	939	7.28
Nominal	0.40	93.35	933	7.21
High	0.40	93.34	891	6.69

Observation: WINT only reduces W_{eff}, so I_{ON} and I_{OFF} fall slightly (939 to 891 μA/μm) while V_{TH} and SS are unchanged.

5.8 Experiment 8: NDEP

Values (low / nominal / high): 1.5×10^{18} / 2.54×10^{18} / 4×10^{18} cm^{-3}

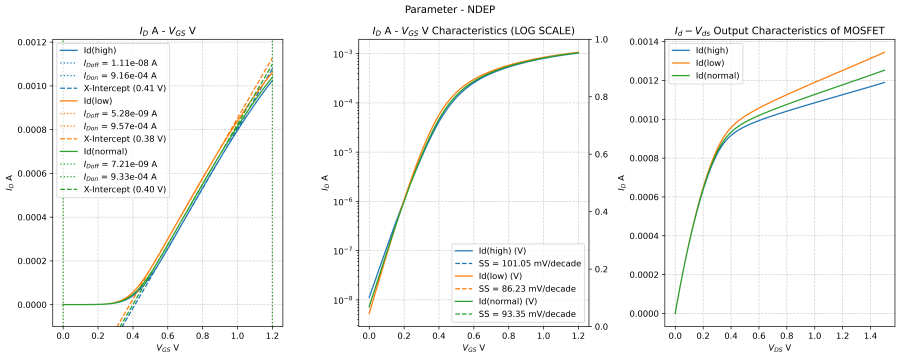


Figure 16: Exp. 8: effect of varying NDEP (left: I_D–V_G with V_{TH}; middle: log scale with SS; right: I_D–V_D).

Case	V_{TH} (V)	SS (mV/dec)	I_{ON} (μA/μm)	I_{OFF} (nA/μm)
Low	0.38	86.23	957	5.28
Nominal	0.40	93.35	933	7.21
High	0.41	101.05	916	11.1

Observation: Higher channel doping increases the depletion capacitance, so SS degrades (86.2 to 101.1 mV/dec), V_{TH} rises slightly and I_{ON} falls through mobility loss.

5.9 Experiment 9: NSD

Values (low / nominal / high): 5×10^{19} / 2×10^{20} / 5×10^{20} cm^{-3}

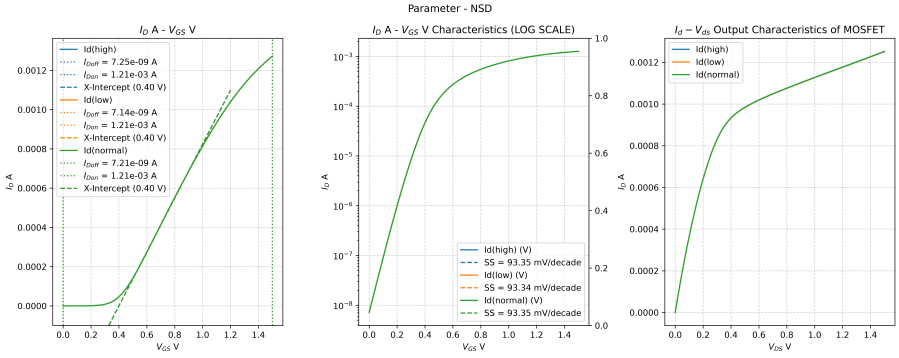


Figure 17: Exp. 9: effect of varying NSD (left: I_D–V_G with V_{TH}; middle: log scale with SS; right: I_D–V_D).

Case	V_{TH} (V)	SS (mV/dec)	I_{ON} (μA/μm)	I_{OFF} (nA/μm)
Low	0.40	93.34	1210	7.14
Nominal	0.40	93.35	1210	7.21
High	0.40	93.35	1210	7.25

Observation: Source/drain doping has a negligible effect on all extracted parameters over this range, since it only enters through the junction and series-resistance models.

5.10 Experiment 10: XJ

Values (low / nominal / high): 10 / 19.6 / 30 nm

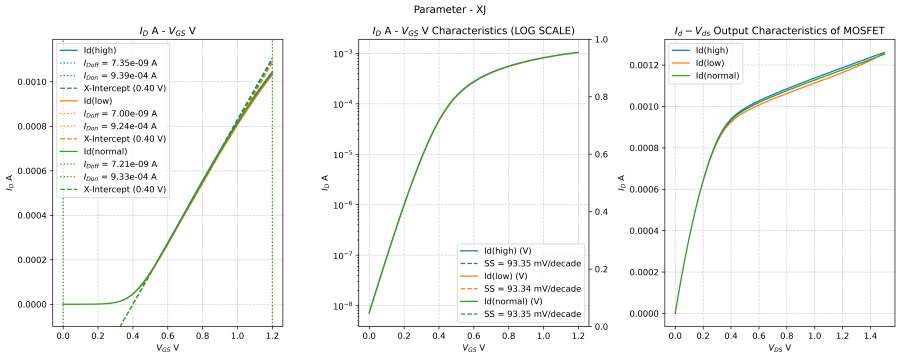


Figure 18: Exp. 10: effect of varying XJ (left: I_D–V_G with V_{TH}; middle: log scale with SS; right: I_D–V_D).

Case	V_{TH} (V)	SS (mV/dec)	I_{ON} (μA/μm)	I_{OFF} (nA/μm)
Low	0.40	93.34	924	7.00
Nominal	0.40	93.35	933	7.21
High	0.40	93.35	939	7.35

Observation: Junction depth has only a small effect: I_{ON} rises slightly (924 to 939 μA/μm) with deeper junctions while V_{TH} and SS stay constant.

5.11 Experiment 11: NGATE

Values (low / nominal / high): 1×10^{19} / 2×10^{20} / 1×10^{21} cm^{-3}

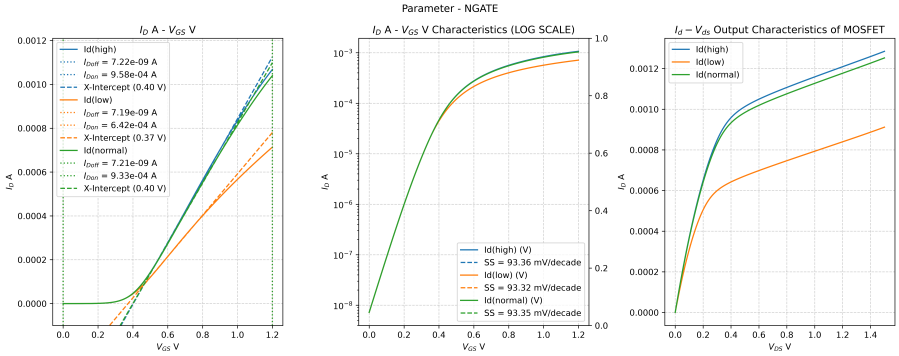


Figure 19: Exp. 11: effect of varying NGATE (left: I_D–V_G with V_{TH}; middle: log scale with SS; right: I_D–V_D).

Case	V_{TH} (V)	SS (mV/dec)	I_{ON} (μA/μm)	I_{OFF} (nA/μm)
Low	0.37	93.32	642	7.19
Nominal	0.40	93.35	933	7.21
High	0.40	93.36	958	7.22

Observation: Low gate doping causes poly-depletion, which acts as extra oxide thickness and cuts I_{ON} to 642 μA/μm, while above 2×10^{20} cm^{-3} the response saturates.

5.12 Experiment 12: U0

Values (low / nominal / high): 300 / 491 / 700 $\mathrm{cm}^2/\mathrm{V}\cdot\mathrm{s}$

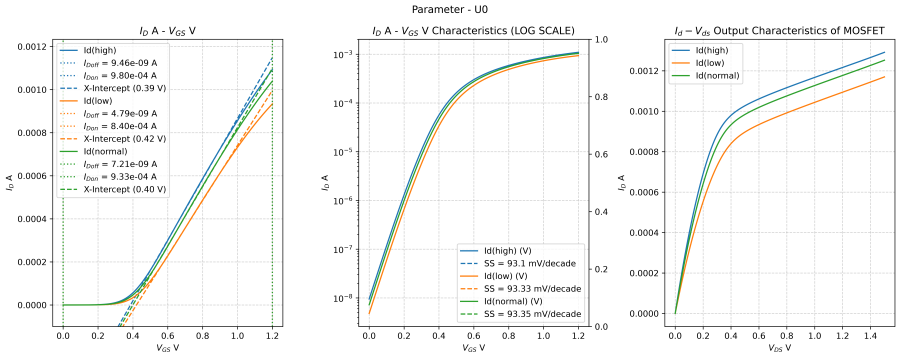


Figure 20: Exp. 12: effect of varying U0 (left: I_D–V_G with V_{TH}; middle: log scale with SS; right: I_D–V_D).

Case	V_{TH} (V)	SS (mV/dec)	I_{ON} (μA/μm)	I_{OFF} (nA/μm)
Low	0.42	93.33	840	4.79
Nominal	0.40	93.35	933	7.21
High	0.39	93.10	980	9.46

Observation: Higher low-field mobility raises I_{ON} (840 to 980 μA/μm) with SS essentially unchanged.

5.13 Experiment 13: VSAT

Values (low / nominal / high): 0.8×10^7 / 1.24×10^7 / 2×10^7 cm/s

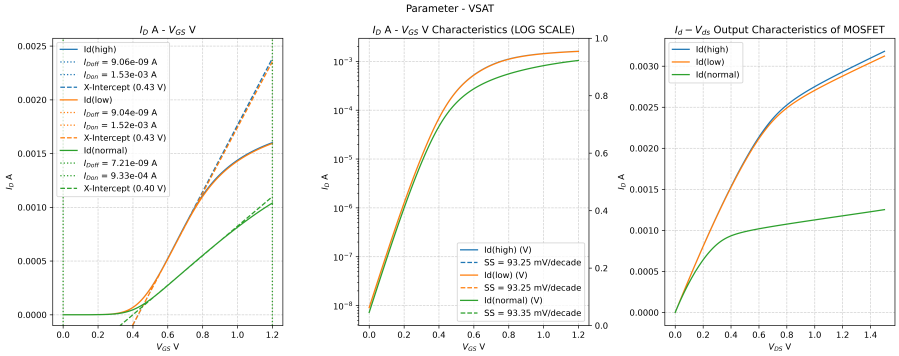


Figure 21: Exp. 13: effect of varying VSAT (left: I_D–V_G with V_{TH}; middle: log scale with SS; right: I_D–V_D).

Case	V_{TH} (V)	SS (mV/dec)	I_{ON} (μA/μm)	I_{OFF} (nA/μm)
Low	0.43	93.25	1520	9.04
Nominal	0.40	93.35	933	7.21
High	0.43	93.25	1530	9.06

Observation: VSAT changes I_{ON} strongly (about 1520–1530 vs 933 μA/μm at nominal) while SS stays near 93.3 mV/dec, showing that it acts only at high lateral field.

5.14 Experiment 14: RDSW

Values (low / nominal / high): 50 / 165 / 400 $\Omega \cdot \mu$m

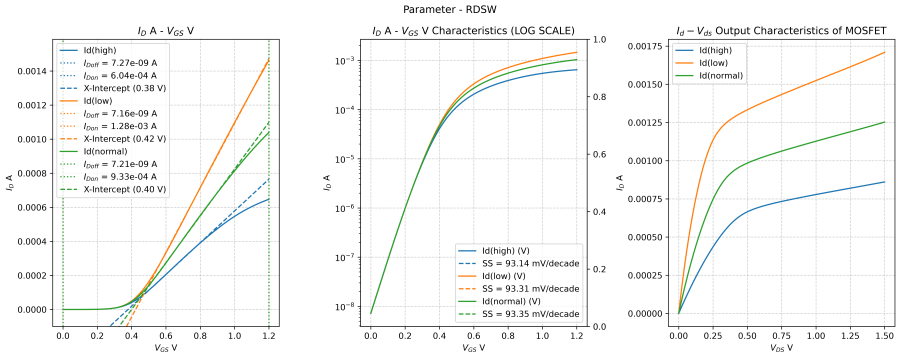


Figure 22: Exp. 14: effect of varying RDSW (left: I_D–V_G with V_{TH}; middle: log scale with SS; right: I_D–V_D).

Case	V_{TH} (V)	SS (mV/dec)	I_{ON} (μA/μm)	I_{OFF} (nA/μm)
Low	0.42	93.31	1280	7.16
Nominal	0.40	93.35	933	7.21
High	0.38	93.14	604	7.27

Observation: A larger parasitic series resistance strongly reduces I_{ON} (1280 to 604 μA/μm) but leaves SS and I_{OFF} unchanged, since it only matters at high current.

Conclusion: V_{TH} is set mainly by VTH0, TOXE, L_{eff} (L, XL, LINT), NDEP and NGATE; I_{ON} is controlled by L_{eff}, U0, VSAT, RDSW and NGATE; SS follows TOXE and NDEP, while W, NSD and XJ have little effect on the normalised characteristics.

6 S6 – Approximate Match to TCAD

Circuit / Testbench

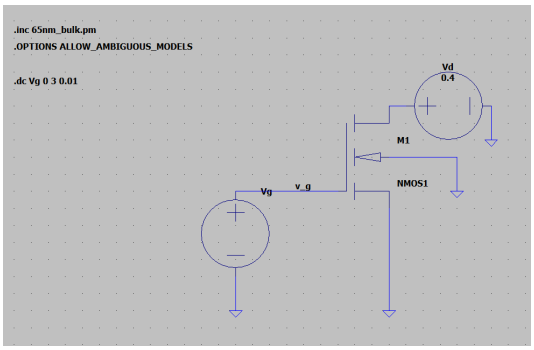
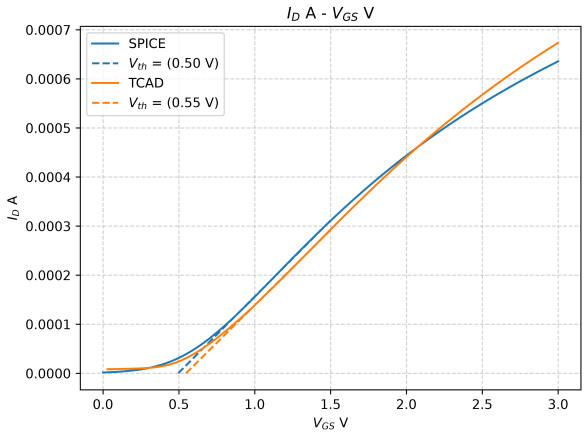
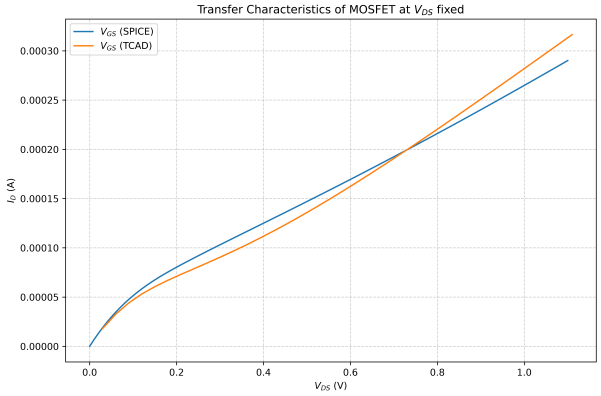


Figure 23: LTspice testbench used for the TCAD comparison.

```
.dc Vg 0 3 0.01/ .dc Vd 0 3 0.01
```



(a) I_D–V_G, $V_{DS} = 0.4$ V (linear)



(b) I_D–V_D, $V_{GS} = 0.9$ V (linear)

Figure 25: TCAD (markers) versus tuned SPICE (lines).

Parameter	PTM value	My value	Reason for change
VTH0	0.423	0.685	To right shift the whole Id-Vg
VOFF	-0.13	-0.6	To pull up the low Voff value in SPICE
Nfactor	1.9	5	For above process and lesser V_{th} shiftand almost equal subthrehold voltage
U0	0.0491	0.02	To pull the currents in saturation region to match I_D-V_G
VSAT	124340	84340	same reason as above
PCLM	0.04	1.1	to increase the slope of the saturation region to match I_D-V_D
ETA0	0.0058	0.0088	to cope up with above changes

Table 6: Parameters modified in `nfit`.

Observation: After tuning, the SPICE lines follow the TCAD markers closely in both I_D–V_G and I_D–V_D, with the threshold region, the saturation current and the output slope all matching.

Conclusion: Shifting VTH0, VOFF and NFACTOR aligns the threshold and subthreshold region, while U0, VSAT, PCLM and ETA0 fit the saturation current and slope, so the PTM card approximately matches the TCAD device.

7 S7 – CMOS Inverter

7.1 Circuit and Parameters

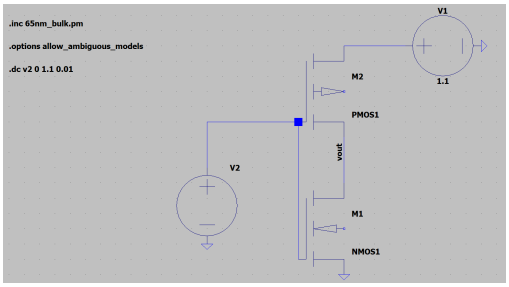


Figure 26: LTspice schematic of the CMOS inverter.

Parameter	nMOS	pMOS
L	65 nm	65 nm
W	1 μm	2 μm
V_{DD}	1.1 V	1.1 V

Table 7: Inverter device parameters.

7.2 Voltage Transfer Characteristic

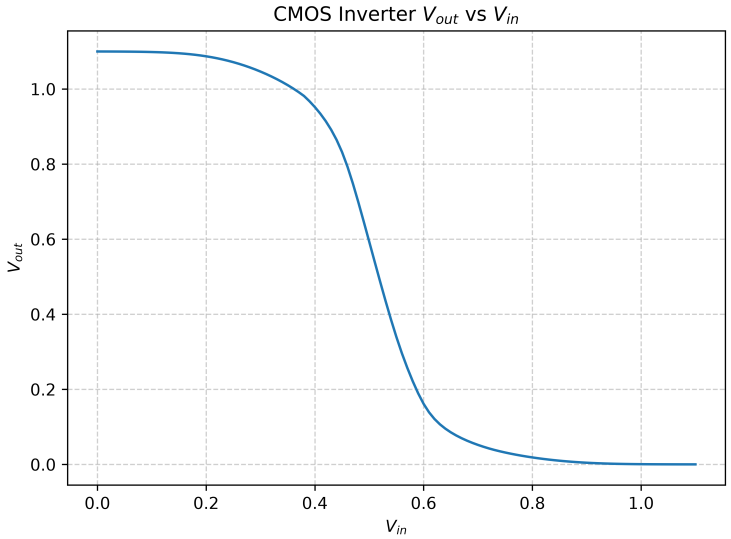


Figure 27: V_{out} versus V_{in} ($0 \rightarrow V_{DD}$) showing V_{OH}, V_{OL}, V_M and the high-gain region.

Device-physics explanation: For $V_{in} \approx 0$ the NMOS is off and the PMOS is on, so the output is pulled to V_{DD} (V_{OH}); as V_{in} nears V_M both transistors are in saturation, the output falls sharply and the gain is maximum; for $V_{in} \approx V_{DD}$ the NMOS is on and the PMOS is off, so the output is pulled to ground (V_{OL}). The output is always the logical complement of the input, which is why the circuit is called an inverter.

Conclusion: The inverter gives full rail-to-rail swing ($V_{OH} = V_{DD}$, $V_{OL} = 0$), and the ratio $W_p/W_n = 2$ compensates the lower hole mobility so that V_M lies close to $V_{DD}/2$, giving a nearly symmetric transfer curve and balanced noise margins.